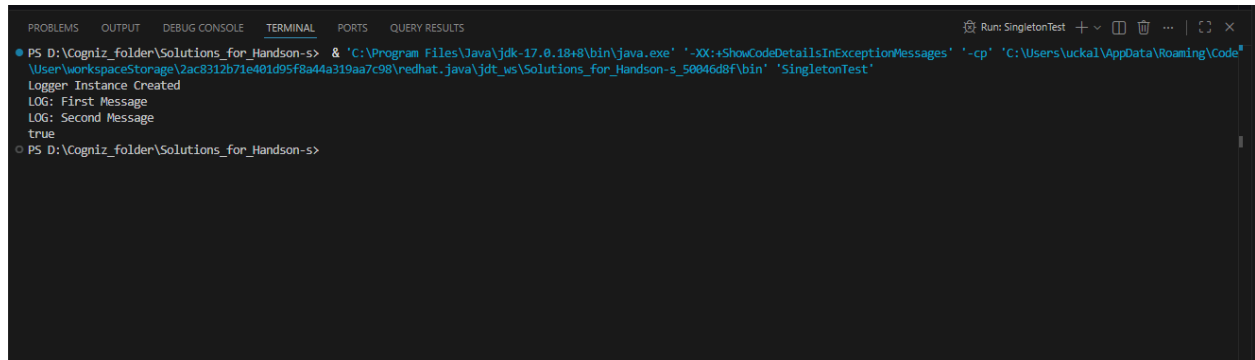


WEEK 1

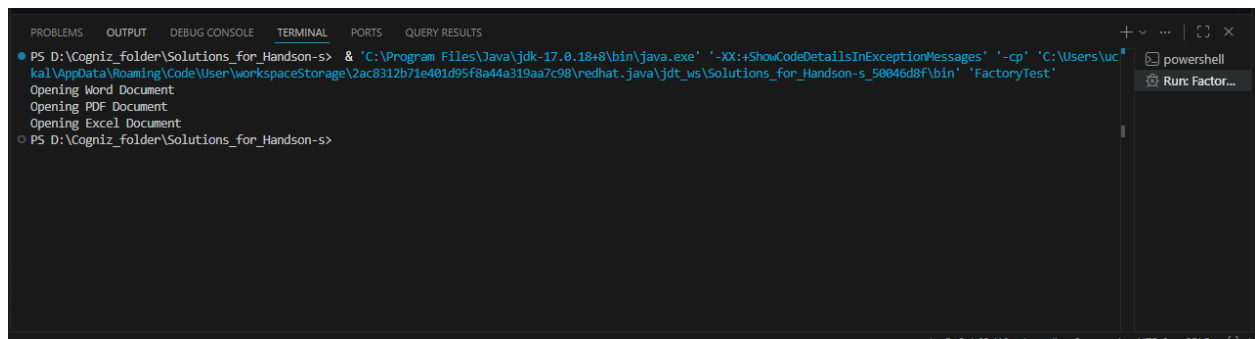
Design Patterns and Principles

Exercise 1: Implementing the Singleton Pattern



```
PS D:\Cogniz_folder\Solutions_for_Handson-s> & 'C:\Program Files\Java\jdk-17.0.18\bin\java.exe' "-XX:+ShowCodeDetailsInExceptionMessages" "-cp" 'C:\Users\uckal\AppData\Roaming\Code\User\workspaceStorage\2ac8312b71e401d95f8a44a319aa7c98\redhat.java\jdt_ws\Solutions_for_Handson-s_50046d8f\bin' 'SingletonTest'
Logger Instance Created
LOG: First Message
LOG: Second Message
true
PS D:\Cogniz_folder\Solutions_for_Handson-s>
```

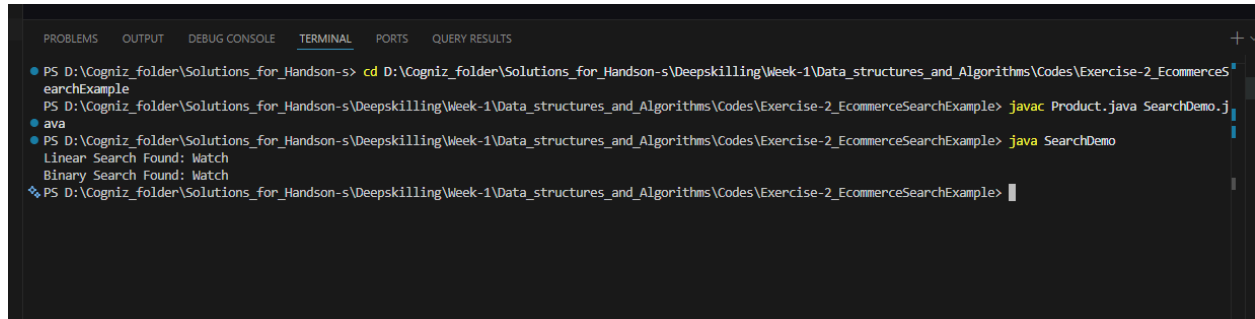
Exercise 2: Implementing the Factory Method Pattern



```
PS D:\Cogniz_folder\Solutions_for_Handson-s> & 'C:\Program Files\Java\jdk-17.0.18\bin\java.exe' "-XX:+ShowCodeDetailsInExceptionMessages" "-cp" 'C:\Users\uckal\AppData\Roaming\Code\User\workspaceStorage\2ac8312b71e401d95f8a44a319aa7c98\redhat.java\jdt_ws\Solutions_for_Handson-s_50046d8f\bin' 'FactoryTest'
Opening Word Document
Opening PDF Document
Opening Excel Document
PS D:\Cogniz_folder\Solutions_for_Handson-s>
```

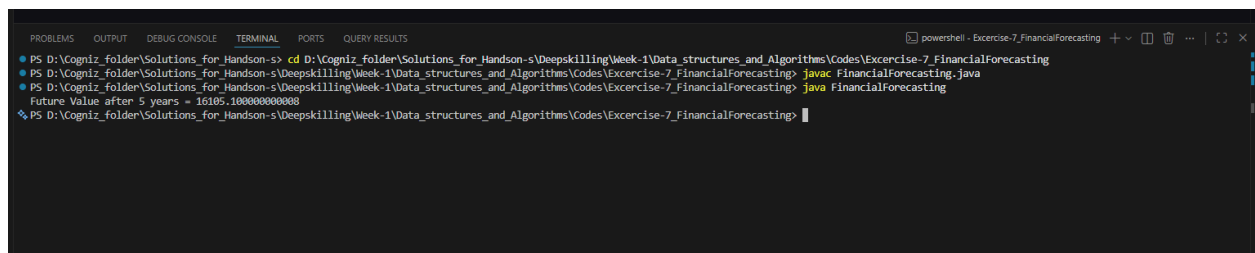
Data Structures and Algorithms

Exercise 2: E-commerce Platform Search Function



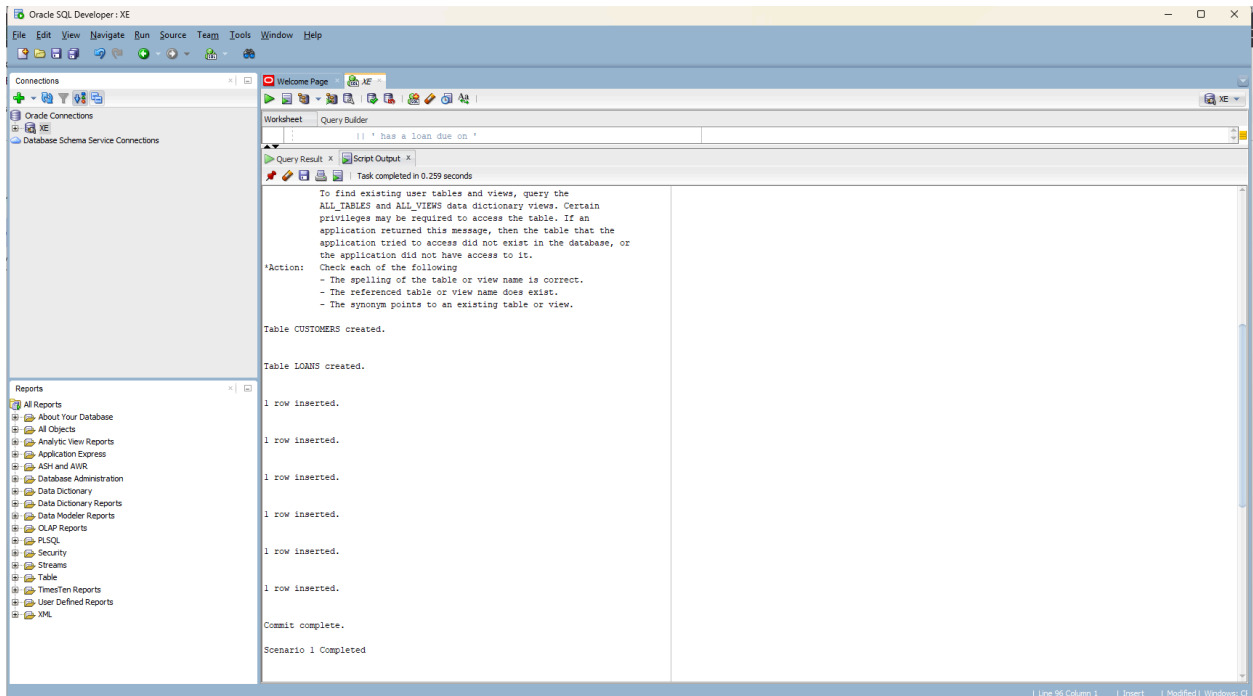
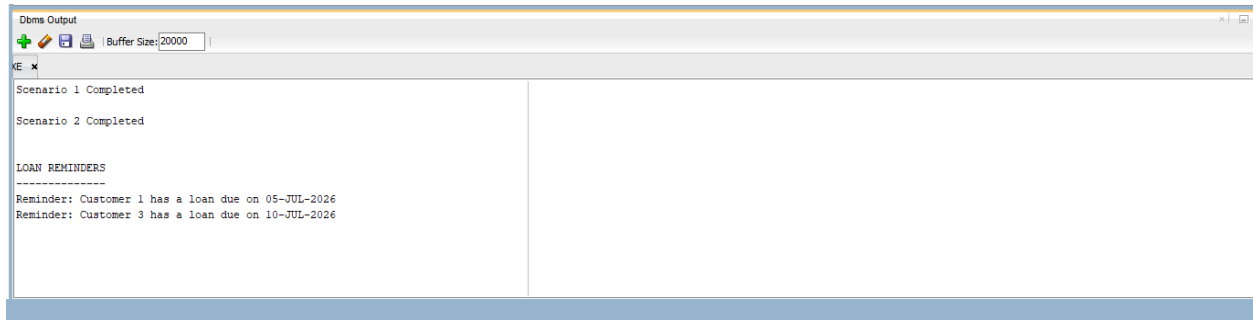
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
PS D:\Cogniz_folder\Solutions_for_Handson-s> cd D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Data_structures_and_Algorithms\Codes\Exercise-2_EcommerceSearchExample
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Data_structures_and_Algorithms\Codes\Exercise-2_EcommerceSearchExample> javac Product.java SearchDemo.java
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Data_structures_and_Algorithms\Codes\Exercise-2_EcommerceSearchExample> java SearchDemo
Linear Search Found: Watch
Binary Search Found: Watch
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Data_structures_and_Algorithms\Codes\Exercise-2_EcommerceSearchExample>
```

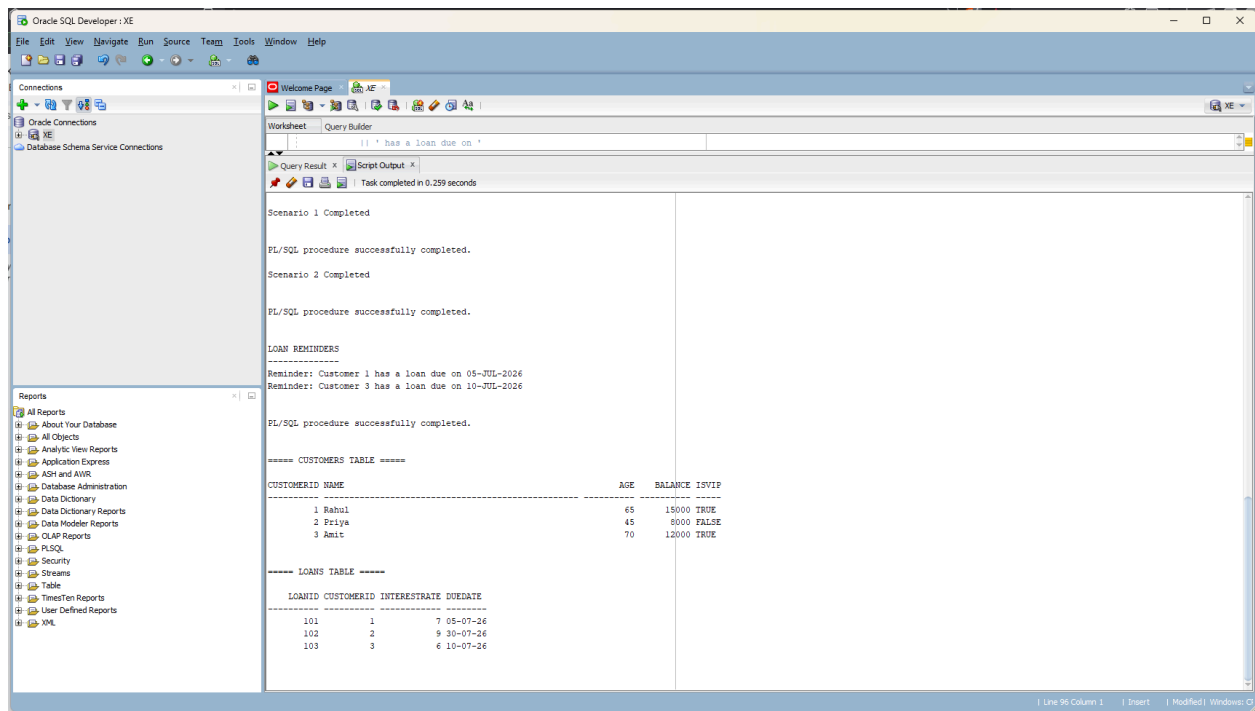
Exercise 7: Financial Forecasting



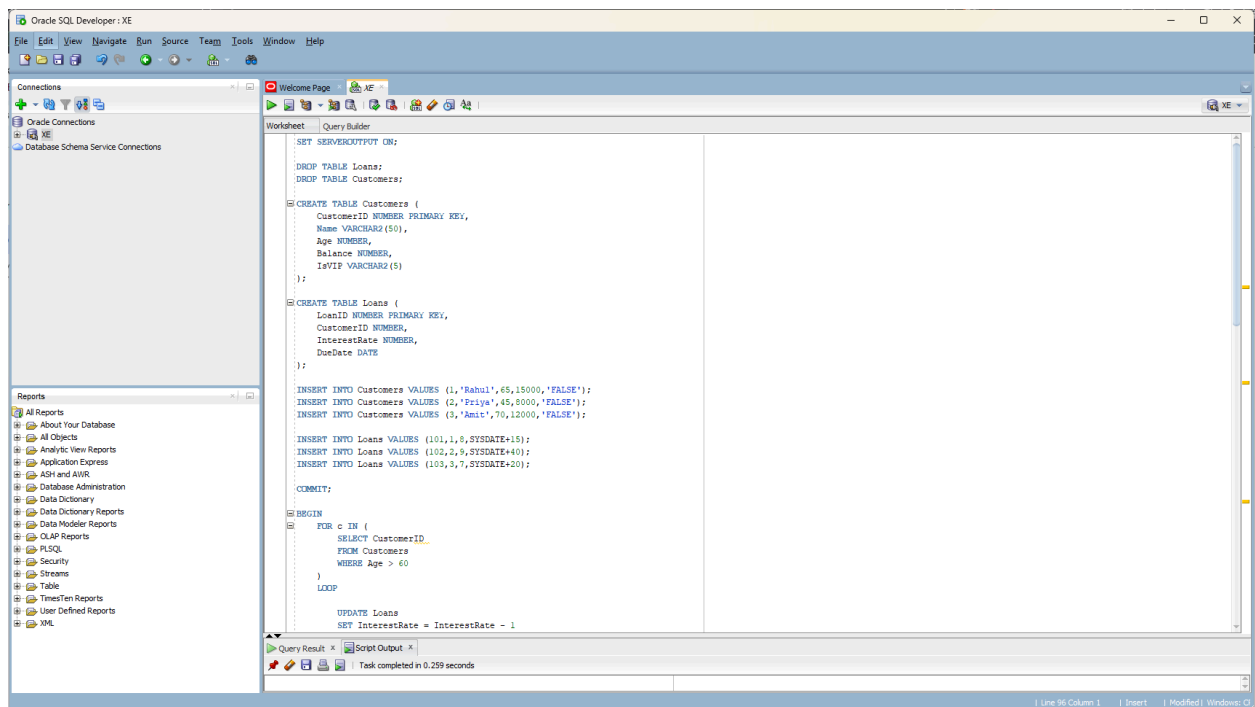
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS powershell - Exercise-7_FinancialForecasting
PS D:\Cogniz_folder\Solutions_for_Handson-s> cd D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Data_structures_and_Algorithms\Codes\Exercise-7_FinancialForecasting
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Data_structures_and_Algorithms\Codes\Exercise-7_FinancialForecasting> javac FinancialForecasting.java
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Data_structures_and_Algorithms\Codes\Exercise-7_FinancialForecasting> java FinancialForecasting
Future Value after 5 years = 16105.100000000008
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Data_structures_and_Algorithms\Codes\Exercise-7_FinancialForecasting>
```

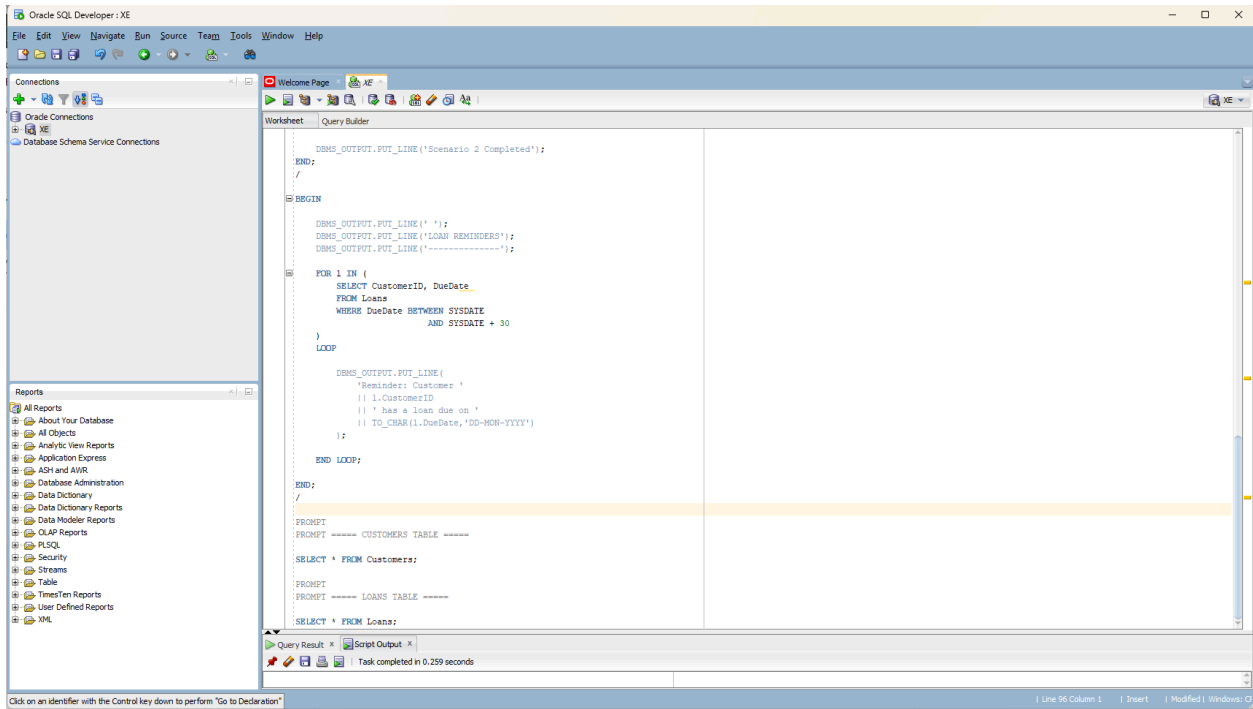
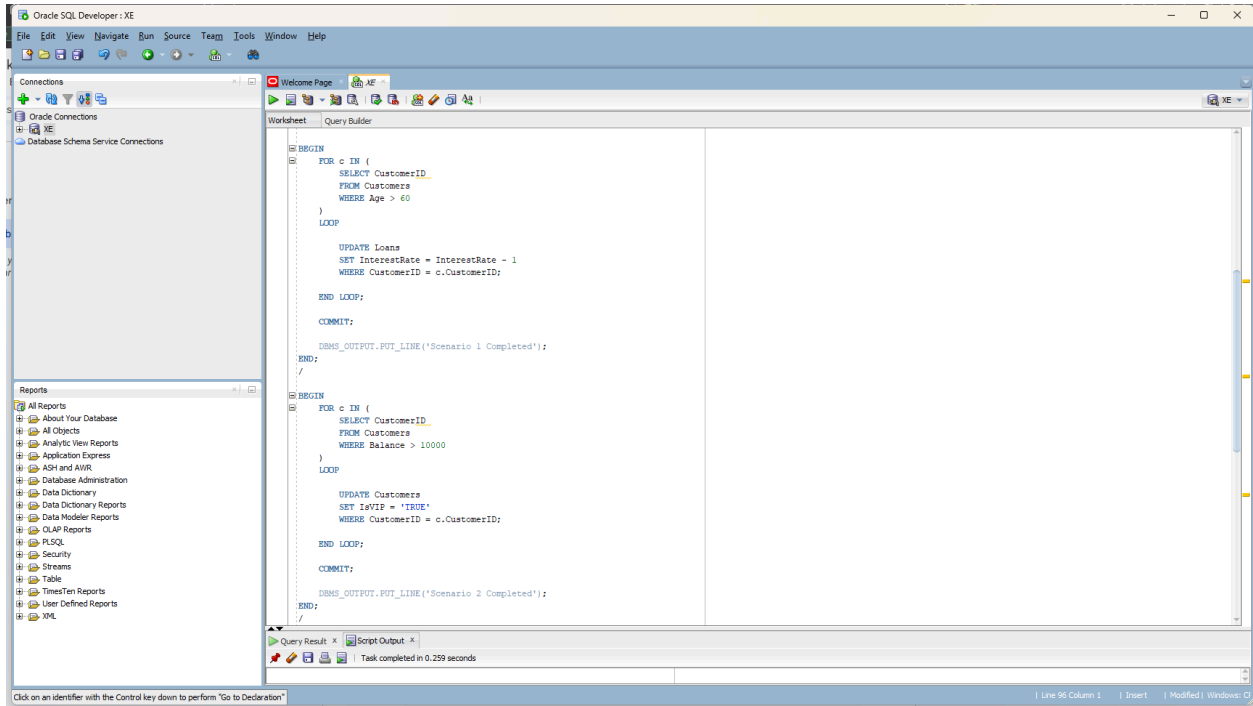
Exercise 1: Control Structures



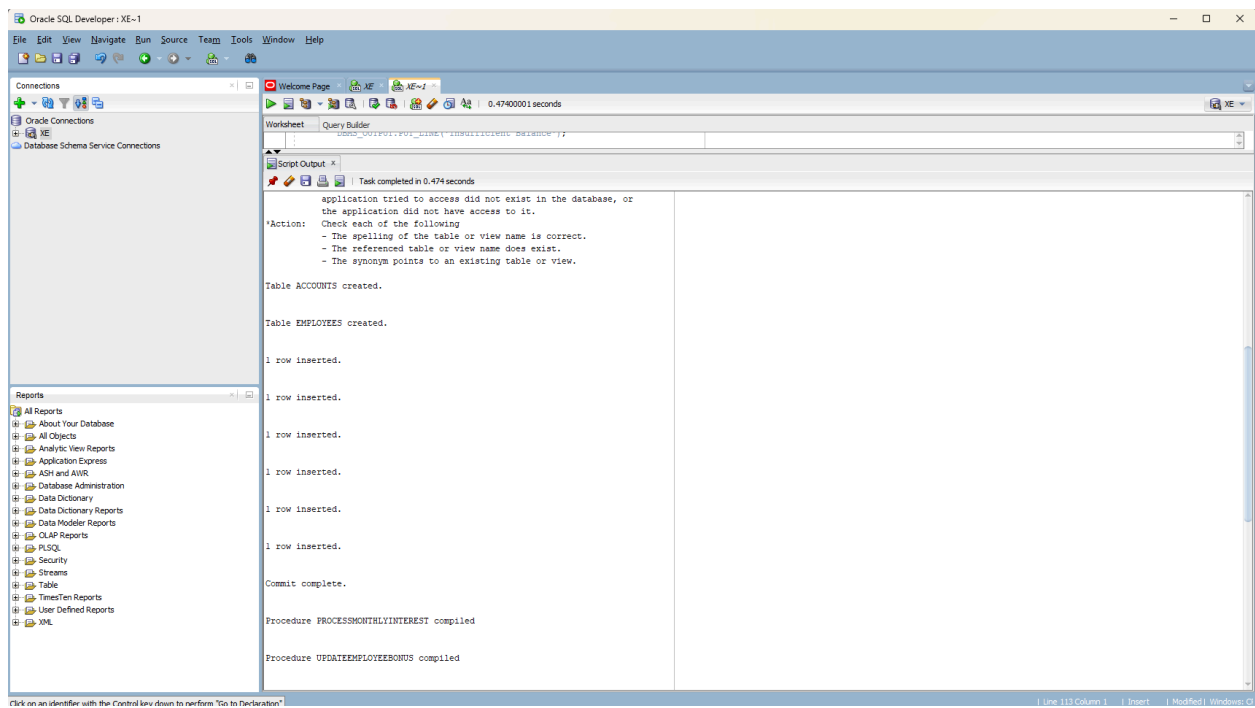
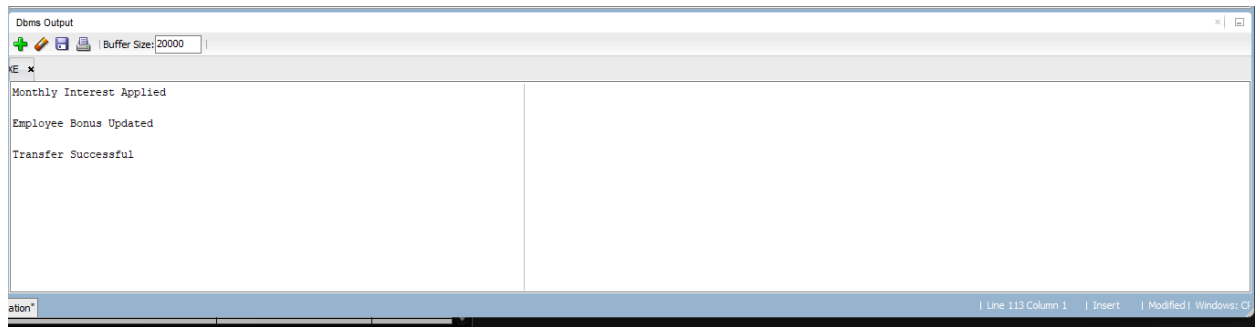


Code





Exercise 3 : Stored Procedures



Task completed in 0.474 seconds

Procedure UPDATEEMPLOYEEBONUS compiled

Procedure TRANSFERFUNDS compiled

Monthly Interest Applied

PL/SQL procedure successfully completed.

Employee Bonus Updated

PL/SQL procedure successfully completed.

Transfer Successful

PL/SQL procedure successfully completed.

==== ACCOUNTS TABLE ====

ACCOUNTID	ACCOUNTTYPE	BALANCE
101	Savings	8100
102	Savings	22200
103	Current	15000

==== EMPLOYEES TABLE ====

EMPLOYEEID	EMPLOYEE NAME	DEPARTMENT	SALARY
1	Rahul	IT	55000
2	Priya	HR	45000
3	Amit	IT	60000

Code -

```

SET SERVEROUTPUT ON;

DROP TABLE Accounts;
DROP TABLE Employees;

CREATE TABLE Accounts (
  AccountID NUMBER PRIMARY KEY,
  AccountType VARCHAR2(20),
  Balance NUMBER
);

CREATE TABLE Employees (
  EmployeeID NUMBER PRIMARY KEY,
  EmployeeName VARCHAR2(50),
  Department VARCHAR2(30),
  Salary NUMBER
);

INSERT INTO Accounts VALUES (101,'Savings',10000);
INSERT INTO Accounts VALUES (102,'Savings',20000);
INSERT INTO Accounts VALUES (103,'Current',15000);

INSERT INTO Employees VALUES (1,'Rahul','IT',50000);
INSERT INTO Employees VALUES (2,'Priya','HR',45000);
INSERT INTO Employees VALUES (3,'Amit','IT',60000);

COMMIT;

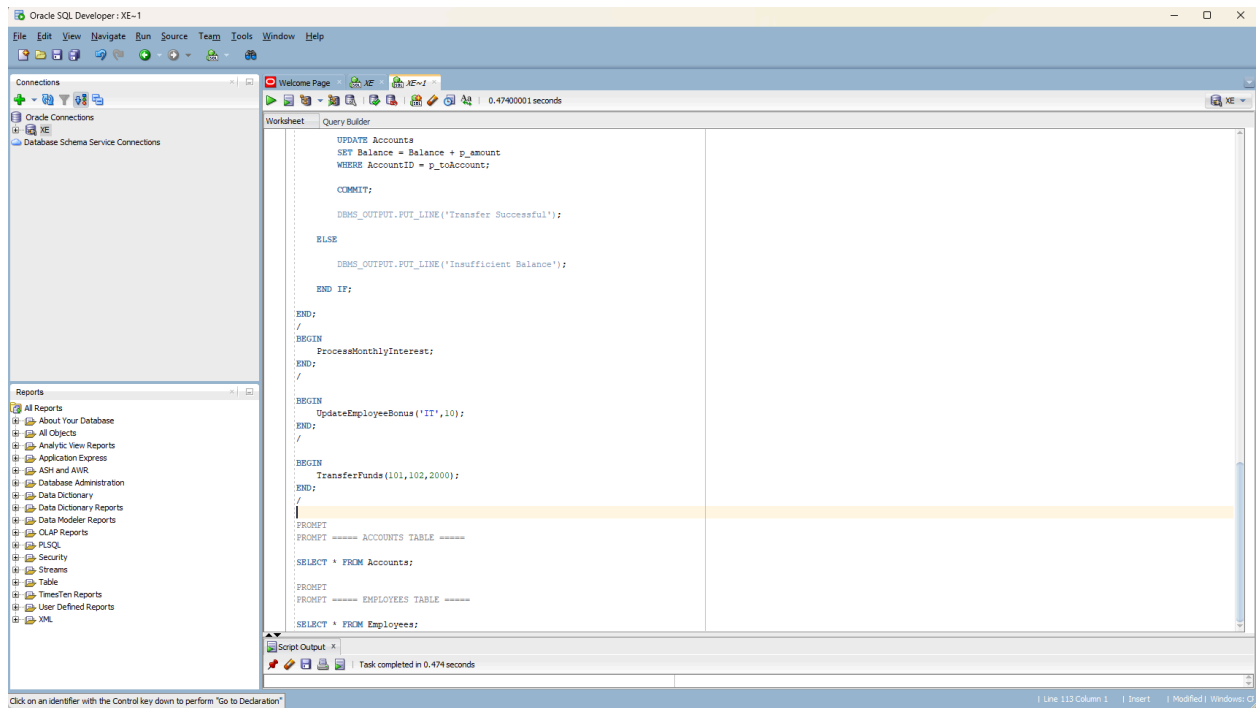
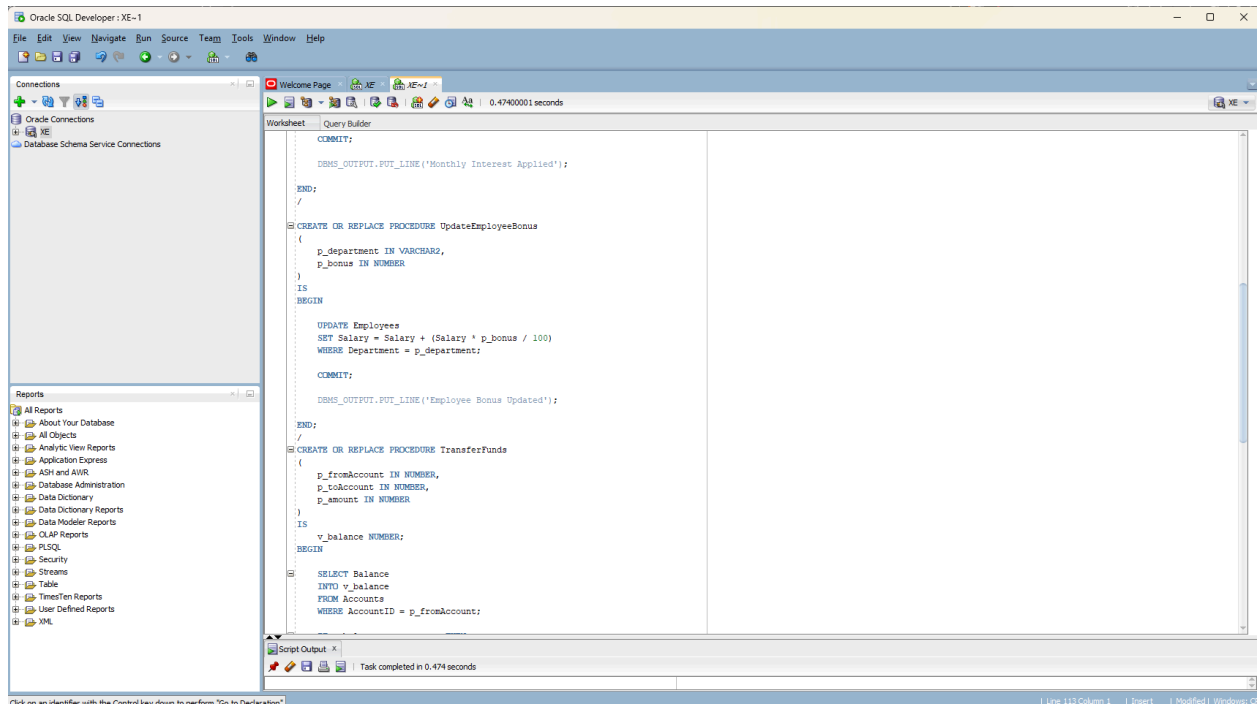
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
IS
BEGIN
  UPDATE Accounts
  SET Balance = Balance + (Balance * 0.01)
  WHERE AccountType = 'Savings';

  COMMIT;

  DBMS_OUTPUT.PUT_LINE('Monthly Interest Applied');

```

Task completed in 0.474 seconds



TDD using JUnit5 and Mockito

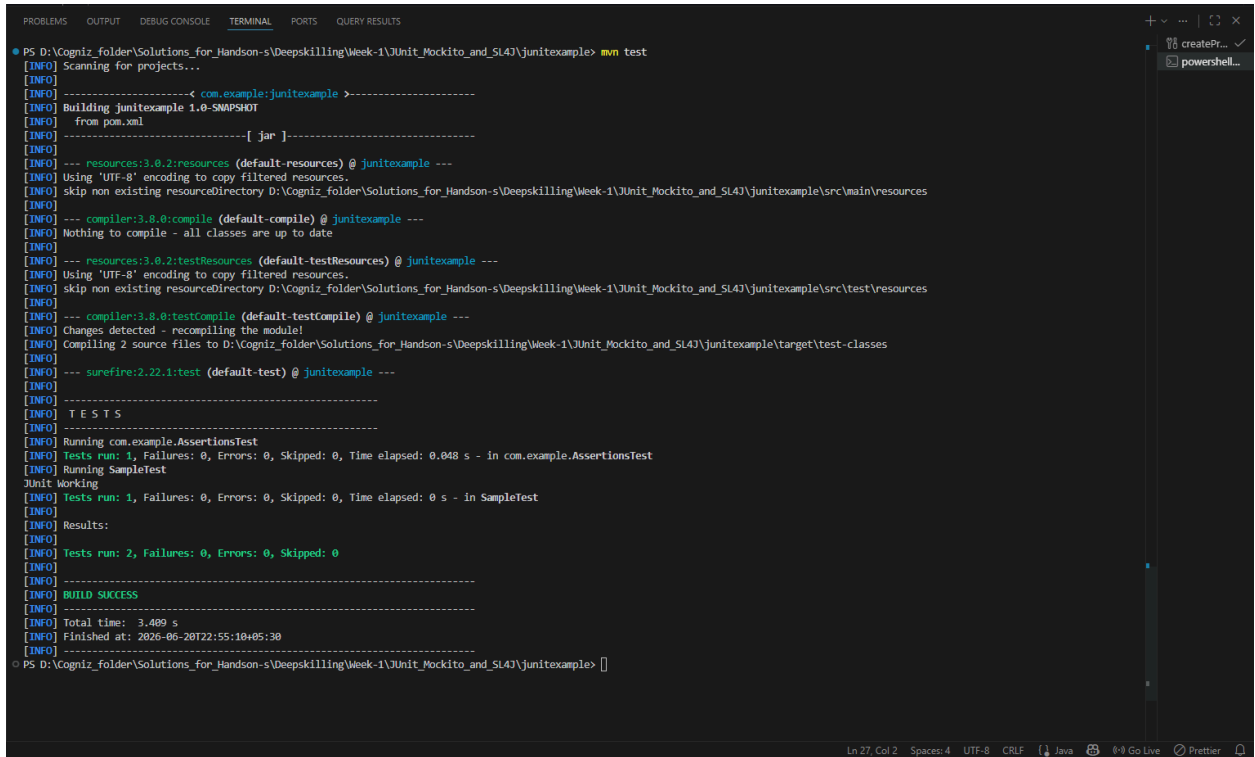
1. JUnit_Basic Testing Exercises

Exercise 1: Setting Up Junit

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit Mockito and SL4J\junitexample> mvn test
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-toolchain/2.2.1/maven-toolchain-2.2.1.jar (38 kB at 13 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/thoughtworks/qdox/qdox/2.0-M8/qdox-2.0-M8.jar (316 kB at 94 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/2.22.1/surefire-junit4-2.22.1.pom (3.1 kB at 19 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/2.22.1/surefire-junit4-2.22.1.jar (85 kB at 214 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/2.22.1/surefire-providers-2.22.1.pom (2.5 kB at 20 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/2.22.1/surefire-providers-2.22.1.jar (85 kB at 214 kB/s)
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running SampleTest
JUnit Working
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.07 s - in SampleTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 22.552 s
[INFO] Finished at: 2026-06-20T22:47:39+05:30
[INFO]
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit Mockito and SL4J\junitexample>
```

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
5   <modelVersion>4.0.0</modelVersion>
6
7   <groupId>com.example</groupId>
8   <artifactId>junitexample</artifactId>
9   <version>1.0-SNAPSHOT</version>
10
11   <name>junitexample</name>
12   <!-- FIXME change it to the project's website -->
13   <url>http://www.example.com</url>
14
15   <properties>
16     <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
17     <maven.compiler.source>17</maven.compiler.source>
18     <maven.compiler.target>17</maven.compiler.target>
19   </properties>
20
21   <dependencies>
22     <dependency>
23       <groupId>junit</groupId>
24       <artifactId>junit</artifactId>
25       <version>4.13.2</version>
26       <scope>test</scope>
27     </dependency>
28   </dependencies>
29
30   <build>
31     <pluginManagement><!-- lock down plugins versions to avoid using Maven defaults (may be moved to parent pom) -->
32     <plugins>
33       <!-- clean lifecycle, see https://maven.apache.org/ref/current/maven-core/lifecycles.html#clean_lifecycle -->
34       <plugin>
35         <artifactId>maven-clean-plugin</artifactId>
36         <version>3.1.0</version>
37       </plugin>
38       <!-- default lifecycle, jar packaging: see https://maven.apache.org/ref/current/maven-core/default-bindings.html#plugin_bindings_for_jar_packaging -->
39       <plugin>
40         <artifactId>maven-resources-plugin</artifactId>
41         <version>3.0.2</version>
42       </plugin>
43       <plugin>
44         <artifactId>maven-compiler-plugin</artifactId>
45         <version>3.8.0</version>
46       </plugin>
47     </plugins>
48   </build>
49 </project>
```

Exercise 3: Assertions in JUnit



```
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:junitexample >-----
[INFO] Building junitexample 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ junitexample ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample\src\main\resources
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ junitexample ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.0.2:testResources (default-testResources) @ junitexample ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample\src\test\resources
[INFO]
[INFO] --- compiler:3.8.0:testCompile (default-testCompile) @ junitexample ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 2 source files to D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample\target\test-classes
[INFO]
[INFO] --- surefire:2.22.1:test (default-test) @ junitexample ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.048 s - in com.example.AssertionsTest
[INFO] Running SampleTest
JUnit Working
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s - in SampleTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.489 s
[INFO] Finished at: 2026-06-28T22:55:10+05:30
[INFO] -----
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample>
```

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

```
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:junitexample >-----
[INFO] Building junitexample 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ junitexample ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample\src\main\resources
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ junitexample ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.0.2:testResources (default-testResources) @ junitexample ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample\src\test\resources
[INFO]
[INFO] --- compiler:3.8.0:testCompile (default-testCompile) @ junitexample ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 3 source files to D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample\target\test-classes
[INFO]
[INFO] --- surefire:2.22.1:test (default-test) @ junitexample ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AAAtest
Setup Method Executed
Teardown Method Executed
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.047 s - in com.example.AAAtest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s - in com.example.AssertionsTest
[INFO] Running SampleTest
JUnit Working
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s - in SampleTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.262 s
[INFO] Finished at: 2026-06-20T23:02:19+05:30
[INFO]
[INFO] -----
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\JUnit_Mockito_and_SL4J\junitexample>
```

3. Mockito exercises

Exercise 1: Mocking and Stubbing

```
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Unit_Mockito_and_SLA7\junitexample> mvn test
[INFO] Scanning for projects...
[INFO] -----[ com.example:junitexample ]-----
[INFO] Building junitexample 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.0.2:resources (default-resources) @ junitexample ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Unit_Mockito_and_SLA7\junitexample\src\main\resources
[INFO] --- compiler:3.8.0:compile (default-compile) @ junitexample ---
[INFO] Nothing to compile - all classes are up to date
[INFO] --- resources:3.0.2:testResources (default-testResources) @ junitexample ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Unit_Mockito_and_SLA7\junitexample\src\test\resources
[INFO] --- compiler:3.8.0:testCompile (default-testCompile) @ junitexample ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 6 source files to D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Unit_Mockito_and_SLA7\junitexample\target\test-classes
[INFO] --- surefire:2.22.1:test (default-test) @ junitexample ---
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AAATest
Setup Method Executed
Teardown Method Executed
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.04 s - in com.example.AAATest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s - in com.example.AssertionsTest
[INFO] Running com.example.MyServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.885 s - in com.example.MyServiceTest
[INFO] Running SampleTest
JUnit Working
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s - in SampleTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.584 s
[INFO] Finished at: 2026-06-21T11:15:54+05:30
[INFO] -----
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-1\Unit_Mockito_and_SLA7\junitexample>
```

```
pom.xml U X ExternalApiJava U MyServiceJava U MyServiceTestJava U AssertionsTestJava U AAATestJava U SampleTestJava U SearchDemoJava U WordDocumentJava U ...
Deepskilling > Week-1 > Unit_Mockito_and_SLA7 > junitexample > pom.xml
3 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5 <modelVersion>4.0.0</modelVersion>
6
7 <groupId>com.example</groupId>
8 <artifactId>junitexample</artifactId>
9 <version>1.0-SNAPSHOT</version>
10
11 <name>junitexample</name>
12 <!-- FIXME change it to the project's website -->
13 <url>http://www.example.com</url>
14
15 <properties>
16 <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
17 <maven.compiler.source>17</maven.compiler.source>
18 <maven.compiler.target>17</maven.compiler.target>
19 </properties>
20
21 <dependencies>
22 <dependency>
23 <groupId>junit</groupId>
24 <artifactId>junit</artifactId>
25 <version>4.13.2</version>
26 <scope>test</scope>
27 </dependency>
28
29 <dependency>
30 <groupId>org.mockito</groupId>
31 <artifactId>mockito-core</artifactId>
32 <version>5.12.0</version>
33 <scope>test</scope>
34 </dependency>
35 </dependencies>
36
37 <build>
38 <pluginManagement><!-- lock down plugins versions to avoid using Maven defaults (may be moved to parent pom) -->
39 <plugins>
40 <!-- clean lifecycle, see https://maven.apache.org/ref/current/maven-core/lifecycles.html#clean_Lifecycle -->
41 <plugin>
```

Exercise 2: Verifying Interactions

```
PS D:\Cogniz_folder\Solutions_for_Handson-s\DeepskillingWeek-1\Unit_Mockito_and_SL4J\junitexample> mvn test
[INFO] Scanning for projects...
[INFO] -----< com.example:junitexample >-----
[INFO] Building junitexample 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.0.2:resources (default-resources) @ junitexample ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\Cogniz_folder\Solutions_for_Handson-s\DeepskillingWeek-1\Unit_Mockito_and_SL4J\junitexample\src\main\resources
[INFO] --- compiler:3.8.0:compile (default-compile) @ junitexample ---
[INFO] Nothing to compile - all classes are up to date
[INFO] --- resources:3.0.2:testResources (default-testResources) @ junitexample ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\Cogniz_folder\Solutions_for_Handson-s\DeepskillingWeek-1\Unit_Mockito_and_SL4J\junitexample\src\test\resources
[INFO] --- compiler:3.8.0:testCompile (default-testCompile) @ junitexample ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 7 source files to D:\Cogniz_folder\Solutions_for_Handson-s\DeepskillingWeek-1\Unit_Mockito_and_SL4J\junitexample\target\test-classes
[INFO] --- surefire:2.22.1:test (default-test) @ junitexample ---
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AAAATest
Setup Method Executed
Teardown Method Executed
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.037 s - in com.example.AAAATest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s - in com.example.AssertionsTest
[INFO] Running com.example.MyServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.874 s - in com.example.MyServiceTest
[INFO] Running com.example.VerifyInteractionTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.006 s - in com.example.VerifyInteractionTest
[INFO] Running SampleTest
JUnit Working
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s - in SampleTest
[INFO] Results:
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.570 s
[INFO] Finished at: 2026-06-21T11:25:02+05:30
[INFO] -----
PS D:\Cogniz_folder\Solutions_for_Handson-s\DeepskillingWeek-1\Unit_Mockito_and_SL4J\junitexample>
```

```
Deepskilling > Week-1 > \Unit_Mockito_and_SL4J > junitexample > src > test > java > com > example > J VerifyInteractionTest.java > VerifyInteractionTest
1 package com.example;
2
3 import org.junit.Test;
4 import org.mockito.Mockito;
5
6 import static org.mockito.Mockito.verify;
7
8 public class VerifyInteractionTest {
9
10     @Test
11     public void testVerifyInteraction() {
12
13         ExternalApi mockApi =
14             Mockito.mock(ExternalApi.class);
15
16         MyService service =
17             new MyService(mockApi);
18
19         service.fetchData();
20
21         verify(mockApi).getData();
22     }
23 }
```

SLF4J logging framework

6. SL4J Logging exercises

Exercise 1: Logging Error Messages and Warning Levels

```
3 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
13 <url>http://www.example.com/</url>
14
15 <properties>
16 <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
17 <maven.compiler.source>17</maven.compiler.source>
18 <maven.compiler.target>17</maven.compiler.target>
19 </properties>
20
21 <dependencies>
22 <dependency>
23 <groupId>junit</groupId>
24 <artifactId>junit</artifactId>
25 <version>4.13.2</version>
26 <scope>test</scope>
27 </dependency>
28
29 <dependency>
30 <groupId>org.mockito</groupId>
31 <artifactId>mockito-core</artifactId>
32 <version>5.12.0</version>
33 <scope>test</scope>
34 </dependency>
35
36 <dependency>
37 <groupId>org.slf4j</groupId>
38 <artifactId>slf4j-api</artifactId>
39 <version>1.7.30</version>
40 </dependency>
41
42 <dependency>
43 <groupId>ch.qos.logback</groupId>
44 <artifactId>logback-classic</artifactId>
45 <version>1.2.3</version>
46 </dependency>
47 </dependencies>
48
49 <build>
50 <pluginManagement><!-- lock down plugins versions to avoid using Maven defaults (may be moved to parent pom) -->
51 <plugins>
52 <!-- clean lifecycle, see https://maven.apache.org/ref/current/maven-core/lifecycles.html#clean_Lifecycle -->
53 <plugin>
54 <artifactId>maven-clean-plugin</artifactId>
```

```
Deepskilling > Week-1 > JUnit_Mockito_and_SL4J > junitexample > src > main > java > com > example > LoggingExample.java > LoggingExample
1 package com.example;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 public class LoggingExample {
7
8     private static final Logger logger =
9         LoggerFactory.getLogger(LoggingExample.class);
10
11     Run | Debug
12     public static void main(String[] args) {
13         logger.error("This is an error message");
14         logger.warn("This is a warning message");
15     }
16 }
17
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS D:\Cogniz_folder\Solutions_for_Handson> & "C:\Program Files\Java\jdk-17.0.18\bin\java.exe" "@C:\Users\uckal\AppData\Local\Temp\cp_hda27gj7842q5e92u0m7j4ly.argfile" "com.example.LoggingExample"
12:02:00.553 [main] ERROR com.example.LoggingExample - This is an error message
12:02:00.556 [main] WARN com.example.LoggingExample - This is a warning message
PS D:\Cogniz_folder\Solutions_for_Handson>
```